

ComPAD in Deepfake Image Detection: Techniques, Comparisons and Challenges

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Abstract. With continual advancements in deep learning, the potential misuse of deep fake is increasing and its detection is in a major scope of work. A model is trained to recognize the patterns in input data, deep fake recognize those patterns in a fabricated way. Sometimes a small, intentional change is added in the data points, these changes are undetectable to humans and confuse the learning model. Those changes are called adversarial perturbations. Compressive adversarial perturbations aim to make those changes even smaller and harder to detect. Authors explore a sophisticated framework - ComPAD (Compressive Adversarial Perturbations and Detection) which is used to detect adversarial attacks. This paper explores the strategies, and provides comparative analysis of methods used by different researchers. Various datasets including UADFV, DeepfakeTIMIT, LFW, FF++, and Deeperforensics are evaluated to achieve the highest metrics. Methods based on convolution neural networks, particle swarm optimization, genetic algorithm and D4 (Disjoint Diffusion Deep Face Detection) are used for detection. Authors also discuss the challenges such as generalization of models across the new data, the continuous evolution of adversarial perturbations that leads to consistent attacks, and the scalability issues for the real time deep fake. Concluding that models can significantly improve the accuracy, robustness and generalization.

Keywords: Deep fake, Adversarial Perturbations, Detection, ComPAD

1 Introduction

Artificial intelligence is a combination of technologies that enables machines to learn, understand and act accordingly. It helps machines to see, translate languages, and analyze data. Virtual assistants like Siri, Alexa, self-driven cars, auto correct and text editors are common examples of AI. Machine learning, computer vision, deep learning, neural network are some commonly used sub-fields of AI.

Deep learning is a part of machine learning, it utilizes neural networks to make computer make decisions in a manner similar to human brain. Neural network comprises of neurons, connections, weights, biases, propagation functions, learning rules. Adaptability, pattern recognition, parallel processing, non-linearity are some advantages of neural networks. One of the key features of Deep learning is that it can handle huge amount of data and it can be trained on unlabeled data as well. Activation function helps in adding non linearity into the network. As the number of layers increases in the network, the understanding of relationships improves.

Advanced techniques like deep learning, generative adversarial networks (GANs) are used being used to digitally manipulate the original content. These technologies are used to develop convincing fake images, audios and videos. The technology and output of the manipulation is together termed as deep fake. Deep learning models are used to create deepfakes. Deepfakes include transforming original content or generating content which is fake. It includes deepfake images, videos and audios. Deepfakes have both advantageous applications and disadvantages as well. Spreading of false information is one of the biggest threat due to deepfakes. Generator and discriminator are the two algorithms used in deepfake technology.

2 Literature Review

The research [1] describes a deep learning-based technique for detecting DeepFake films that identifies artifacts from face-warping operations, which are required owing to the restricted resolution of produced pictures. This approach uses convolutional neural networks (CNNs) to capture these distinguishing artifacts, removing the requirement for massive datasets for training. When applied to video authenticity verification, it was evaluated on the UADFV and DeepfakeTIMIT datasets, and the ResNet50 model beat cutting-edge techniques such as Two-stream NN, MesoNet, and HeadPose by up to 21%. This technique is more resilient across DeepFake sources, with future enhancements focused on video compression robustness and specialized network topologies.

In this study [2], it represents model uncertainty owing to unknown dependency. The authors explore the ordering relations between aggregation sets impacted by distribution mixes and quantile mixtures of marginals. The study discovers that homogenizing marginals using these mixes typically increases the aggregation set size, magnifying model uncertainty; however, this impact is more obvious with distribution mixtures than with quantile mixtures. They also investigate the worst-case values of risk

measurements such as Value at Risk (VaR), focusing on marginal distributions with monotone densities, and find relevant inequalities. Numerical results back up the theoretical conclusions, leading to conjectures and practical applications in portfolio diversification with dependency uncertainty and merging p-values in multiple hypothesis testing. The study provides important insights on the influence of marginal and dependency uncertainty on robust risk aggregation, pertinent to financial risk management.

The authors [3] address the challenges of deep face detection with deep learning techniques enhanced by hybrid optimization methods such as particle swarm optimization (PSO) and genetic algorithms (GA). These strategies optimize neural network weights and parameters to achieve robust detection models. The system is assessed on the Proteus platform, demonstrating the performance of hybrid optimization using measures such as accuracy, precision, recall, and F1 score. Parallel and distributed computing is enabled for real-time detection and improving scalability. The combination of PSO and GA creates a clear, efficient, and resilient solution for content security.

The evaluated research [4] looks on the vulnerability of deep neural networks (DNNs) to adversarial instances, which are minimally modified inputs that can fool DNNs while staying undetected to humans. The article categorizes ways to producing hostile instances and discusses their applications and defenses. The key phases include assessing the robustness of DNNs, offering a taxonomy of adversarial example generation methods, and emphasizing the necessity for a benchmark platform to standardize attack and defense evaluations. The authors emphasize the continuous "arms race" between adversarial assaults and defenses, as well as the importance of strong assessment procedures in ensuring DNNs dependability in real-world contexts.

This article [5] examines recent technical breakthroughs. It focuses on the key four sub-fields: 1) face swapping. 2) Face recreation. 3) Creating talking faces, and 4) Editing facial attributes. This research also demonstrates how the approach evolved from VAE (variational Autoencoders) to GAN (generative adversarial network) and the progressive transition from single forward GAN models to multi-step diffusion models. The framework is separated into three parts: the generating framework (VAE, GANs, and diffusion), the discriminative neural network (CNNs, transformers), and the Neural Radiance Field. Deep fake detection uses datasets such as LFW, CelebA, VGGFace, and FFHQ, whereas forgery detection uses UADFV, TIMIT, FF++, Deeperforensics-1.0, and DFDCP.

The paper [6] addresses the challenge of recognizing diffusion-generated deep fake images, especially when adversaries utilize undetectable perturbations to prevent detection. D4 enhances detection robustness by including an ensemble of models that operate over discontinuous frequency spectrum subsets, making it more resistant to adversary manipulation. The key finding is to leverage frequency domain redundancy and saliency partitioning to distribute frequency components over many models, hence reducing the effective input subspace where adversarial deep fakes may be optimized.

The ability to discern between actual and fraudulent information is becoming increasingly important as the number of digitally changed photographs grows. DenseNet121 [7] is used in this study on a chosen dataset of actual and false photos, resulting in high accuracy (93.42%) and good metrics including precision, recall, and AUC. DenseNet121's design successfully catches tiny details by recycling features, making it

ideal for identifying subtle variations between original and changed pictures. Grad-CAM is also used to emphasize portions of the picture that had the most effect on the model's judgments, increasing transparency, critical for media verification applications.

Furthermore, the Vision Transformer (ViT) [8], which was initially designed for word activities, is now employed to analyze images by focusing self-attention on tiny patches. ViT enriches inputs by combining patch embedding's with CNN features from a pre-trained model. A notable element discussed in [8] is the addition of a distillation token to increase performance by learning from a teacher network (the pre-trained model). During training, the model is driven by both the teacher network and the distillation tokens; in testing, the distillation token replaces the class token, resulting in better performance on real-world datasets, improving the accuracy.

Conventional methods of detection use labeled dataset for training focusing on low level anomalies and not generalizing well on unseen data. Self-supervised contrastive learning [9] allows models to learn from unlabeled data. High level visual-representations extraction can be done, improving resilience to common image distortions and generalization. Graph Convolutional Networks (GCNs) help in capturing complex inter-dependencies by capturing relationships between different regions of image. Transformers, however help in understanding the long range dependencies. Combining GCNs and transformers help in understanding of local and global patterns and improving accuracy, resilience and generalization.

3 Proposed Methodology

This paper highlights recent improvements in deep face identification and adversarial perturbation defenses. We selected research from credible resources such as arXiv, ResearchGate, and computer vision conferences. The selected publications shed fresh light on convolutional neural networks (CNNs) [1] [13], hybrid optimization approaches [3], and novel frameworks like ComPAD and D4 [5]. We chose papers that span a variety of approaches and datasets to offer a comprehensive overview of current research in this field.

3.1 Different methods and technologies used

This section goes over the key approaches used in the evaluated studies to combat deep fake manipulation and adversarial attacks on DNNs.

ComPAD Framework: ComPAD, or Compressive Adversarial Perturbations [10] and Detection, is a framework for detecting deep fake pictures that have undergone modest but purposeful modifications. ComPAD uses compressive methods to detect very minor disturbances while maintaining picture quality. ComPAD improves the model's resilience against adversarial assaults by lowering perturbation size and increasing detection sensitivity. ComPAD identifies compressive adversarial perturbations successfully, however its scalability may be limited when used to high-resolution movies due to processing constraints. Its emphasis on small alterations makes it perfect

for identifying slight perturbations, but it may be less successful against large-scale picture modifications.

CNN-based Deepfake Detection: CNNs are widely used in deepfake detection due to their ability to capture minute details, such as face-warping artifacts [1] like misaligned facial geometry or resolution mismatches. One notable study utilized the ResNet50 model to identify artifacts within the UADFV and DeepfakeTIMIT datasets. In this, the negative example generation is simplified by simulating the affine face wrapping step. These negatives are then used in training to recognize realistic distortion. This method achieved up to 21% higher accuracy than previous approaches, showcasing CNNs' [13] capability to detect deep fakes with high precision.

Hybrid Optimization Techniques (PSO and GA): To increase model efficiency and robustness, numerous researchers coupled particle swarm optimization (PSO) and genetic algorithms (GA) with neural network training. This combination optimizes network weights and parameters, resulting in improved detection performance metrics including accuracy, precision, and recall. These hybrid systems, which employ platforms like Proteus, offer a scalable option for real-time applications [3]. Combining PSO and GA with neural network models improves real-time identification, especially for large multimedia datasets. However, these hybrid models usually need significant processing power, which may restrict their application in resource-constrained environments.

Disjoint Diffusion Deep Face Detection (D4) [6] is a technique that employs disjoint ensemble models to extract and analyze various frequency spectrums, making it more resistant to adversary perturbations. This technique uses frequency redundancy to selectively divide frequency components among many models, limiting the input space that adversarial deep fakes may exploit. The D4 technique has proven to be extremely robust against black-box adversarial attacks, as well as resilient over a wide range of datasets. D4 is extremely resistant to adversarial attacks because of its use of fractured frequency spectrum ensembles. This technique has proved the ability to generalize across unknown data distributions, although the need for several models may result in greater computing cost during implementation.

ViT (Vision Transformer): ViT uses self-attention technique [8] after breaking the image into smaller patches, which allows to focus simultaneously on different parts of the image. The patches are converted into numerical representations, i.e. a vector. These patch embedding's are combined with CNN features (pre trained model). Patch embedding's and CNN features are the two inputs to the ViT model. Combination of CNN features and patch embedding's providing a more explanatory input to the model. Incorporating distillation and class tokens in it, providing a deepfake detection mechanism with good accuracy of results.

Self-supervised graph transformers: With the help of self-supervised contrastive learning [9], models can learn through unlabeled data. Combination of self-supervised learning technique, GCNs, transformers and relevancy maps provide the features of generalization, better understanding of relationships among different parts of image and better transparency in the detection mechanism.

3.2 Different methods and technologies used

Generalization across Deep Fake Sources: Models like CNNs are trained on specific datasets and may not generalize effectively to different types of deep fake content, especially with the rise of new generation techniques. This limits their robustness in real-world applications, where they may encounter diverse and unseen fake content.

Scalability for Real-Time Detection: Many existing techniques particularly the ones based on hybrid optimization require significant processing power. That makes them computationally efficient but they can't meet the requirements for low-latency and high throughput scenarios. This poses an issue in real time detection of deep fake.

Adversarial Attack Resilience: With the attackers using sophisticated methods, the adversarial perturbations continue to evolve. Techniques like D4's disjoint ensemble approach show promise, but adversarial attacks remain a persistent threat that requires continual adaptation of detection models.

Benchmark Standardization: The lack of a unified benchmark for testing both attack and defense strategies limits the comparability of different models. Consistent benchmarking would facilitate objective evaluation and improvement across studies, enabling more robust detection frameworks.

4 Results and Discussion

Table 1. Summary of Deepfake Detection Methods and Their Performance.

Detection Method	Strengths	Limitations
ComPAD	High detection for minor perturbations	Computational demand restricts large-scale use.
CNN (ResNet50)	Strong artifact detection performance	Needs large training datasets; may struggle with compression
PSO & GA Hybrid Optimization	High detection accuracy, good scalability	High power and processing requirements
D4 Disjoint Ensembles	Resilient to adversarial attacks, robust across datasets	High complexity and processing requirements.
ViT (Vision Transformer)	Better detection of fake images and increased accuracy	High computational demand for training and inference
Self-supervised graph transformer	Improved generalization and transparency. Better understanding of inter-dependencies of different regions of an image	Flexibility issues in graph generation

In one of the research papers [1], a deep learning method, convolutional neural networks, is used for detecting Deep Fake films by capturing the distinguished artifacts. It is inferred that their ResNet50 model outperforms Two-stream NN by 16%, Meso-4 by

17% and MesoInception-4 by 21% when evaluated on UADFV and DeepfakeTIMIT datasets.

Table 2. Comparison of different techniques.

Title	Authors	Year	Techniques	Dataset	Findings
Exposing DeepFake Videos By Detecting Face Warping Artifacts	Yuezun Li, Siwei Lyu	2018	CNNs, Res-Net50	UADFV, DeepfakeTIMIT	CNN-based detection with Res-Net50 outperformed other models in artifact detection by up to 21%.
Ordering and Inequalities for Mixtures on Risk Aggregation	Yuyu Chen, Peng Liu, Yang Liu, Ruodu Wang	2020	Marginal Distributions	NA	Analyzes risk and uncertainty in DNN robustness against adversarial perturbations
Development of Deepfake Detection Techniques for Protecting Multimedia Information using Deep Learning	Siva Rama Lingham N Et al.	2024	PSO and GA, Hybrid Models	Custom Platform (Proteus)	Demonstrates faster convergence with PSO-GA hybrid models, improving accuracy and recall for detection.
Deepfake Generation and Detection: A Benchmark and Survey	Gab Pei Et al.	2024	VAE, GAN, Diffusion Models	LFW, FF++, UADFV	Discusses detection performance of VAEs, GANs, and diffusion models across multiple datasets.
D4: Detection of Adversarial Diffusion Deepfakes Using Disjoint Ensembles	A Hooda Et al.	2024	D4, Disjoint Ensembles	FF++, Deeperfomics	Demonstrates robustness to adversarial attacks by using frequency spectrum partitioning.
Detecting AI-generated images with CNN and Interpretation using Explainable AI	BM Gurusamy Et al.	2024	Dense-Net121, (Grad-CAM) technique	extensive dataset of ‘real’ and ‘fake’ images	Gives an accuracy of 93.42, helps in distinguishing between real and fake content, hence decision making

Self-Supervised Graph Transformer for Deepfake De- tection	Aminollah Khormali; Jiann-Shiun Yuan	2024	self-super- vised learn- ing , GNNs, transformers , relevancy maps	FF++, Ce- leb- DF(V2), DeeperFo- rensics, FaceShifter ,DFDC	Improved accu- racy and general- ization. Better deepfake detec- tion, improved transparency and robustness.
Deepfake Detection Scheme Based on Vision Transformer and Distillation	Young-Jin Heo, Young-Ju Choi, Young- Woon Lee, Byung-Gyu Kim	2021	CNN fea- tures and patch-based positioning model	Deepfake Detection (DFDC) Dataset	Obtains 0.978 of AUC and 91.9 of f1 score, making it a better model than SOTA model.

The paper [2] observes the relationship between aggregation sets impacted by distribution mixes and quantile mixtures of marginals. It also investigates the worst case values of risk measurements such as VaR (Value at Risk). This study provides insights on the influence of marginal and dependency uncertainty on robust risk aggregation.

The uncommon method of using particle swarm optimization, genetic algorithms, and importance of ensemble models are discussed and are more efficient in accumulation of diverse deep fake detection [3]. Performance metrics like accuracy, F1 score, AUC, precision and recall have confirmed the same.

The paper [4] discusses about the adversarial instances and how they affect the deep neural networks. Concluding that these instances poses a risk and are undetected to humans. The paper, assesses the robustness and the necessity for benchmarking for standardization of attack and defense.

The paper [5] tells that GAN based models surpasses the traditional image synthesis and produce more realistic images. Detection, however, suffers especially when the images are generated with attention mechanisms. The article focuses on the common four sub fields including face swapping, face recreation, talking faces and editing facial attributes.

One of the papers discusses an ensemble approach to deepfake detection called the Disjoint Diffusion Deepfake Detection (D4) [6], which exploits redundancy in frequency space by saliency partitioning. It offers robustness and lowers attack success rates under the black-box attacks.

A popular deep learning model - DenseNet121 is used on a dataset of 'real' and 'fake' images to create an AI model for image identification [7]. It shows an impressive accuracy of 93.42, and other metrics such as precision, recall, and area under the ROC curve. Gradient-weighted class activation mapping (Grad-CAM) is employed to enhance the transparency and interpret-ability of the model.

In the paper [8], a method is designed such that it combines the CNN features and patch based positioning model to interact with all positions, and for solving false negative issues. Dataset of Deepfake Detection (DFDC) is used and giving an AUC score of 0.978 and F1 score of 91.9 which better than the previous state-of-the-art model, under the same parameters.

The paper [9] presents the technique of self-supervised graph transformer. It is a combination of self-supervised contrastive learning, GCNs, transformers and relevancy maps. Generalization is improved with self-supervised contrastive learning, GCNs and transformers help in understanding inter-dependencies among different regions of an image, and relevancy maps improve transparency.

Faceforensics++, DFDC are some of the benchmark datasets for deep fake detection and XceptionNet, MesoNet, VGG-Face, ResNet are some of the benchmark models for deep fake detection. Accuracy, computational cost and real time performance are also important benchmark criteria's. Deepfake detection models have a high computational cost, especially in real-time performance, which means the system should process in a matter of milliseconds. Based on the accuracies provided, ComPAD has a moderate computational cost.

Sensitive data like facial features and body movements are being processed during the detection process. Researchers should ensure that deep fake technologies are developed in such a way that it handles ethical issues and privacy concerns. Models should also be made to be unbiased towards underrepresented groups of society, and abide by the ethical norms. Deepfake is also a potent tool to manipulate opinions and to tarnish images.

5 Summary and conclusions

In this work, authors performed a comparison between different proposed techniques of deepfake detection. One of the major concerns regarding deepfake technology is spreading of false information, digital data from different platforms is manipulated in order to regenerate or generate fake images, audios and videos. One of the ways mentioned used convolutional neural networks (CNN) for detection, which reduces the requirement for big picture training datasets. The ResNet50 model was utilized for video authenticity checking. Another detection approach used particle swarm optimization (PSO) and genetic algorithms (GA), which optimized weights and parameters to produce robust models. Vulnerability of deep neural networks (DNNs) to adversarial attacks is discussed along with the discussion for need of standardized benchmark platforms and robust assessment processes. One of the paper proposes the Combination of generative models (VAE, GANs, and Diffusion) along with discriminative models and neural radiance fields as a method for deepfake detection D4: Disjoint Diffusion Deep Face Detection addresses the challenge of identifying diffusion generated fake images. This method uses ensemble technique for enhancing the capability of the detection model. In this paper we infer that, use ensemble techniques that is combining different models for detecting fake data will lead to improved accuracy. Combining generative models, discriminative models will provide a better differentiation between real and fake data.

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